

Iran War Risk 2026 — Sensitivity of Global Financial Variables

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Replication of Rigobon & Sack (2003), *The Effects of War Risk on U.S. Financial Markets*, applied to the **2026 Iran war** (window **28 Feb 2026** → **present**). The unobservable war-risk factor is recovered with **identification-through-heteroskedasticity** (Rigobon 2003), using **NLP on war news** (GDELT volume + FinBERT + a war lexicon) to flag high/low news days — the paper's hand-read newspaper list, automated.

1. Task and approach

The professor's brief: "Use NLP on war news to flag high-news days (1 = high, 0 = low), then apply heteroskedasticity-based identification. The war-risk effect comes from the shift in variance on those days, so the direction of each story doesn't need to be known." Deliverables: replicate the paper's three tables for the Iran war (28 Feb → present), and evaluate whether heteroskedasticity-based identification is the best approach.

Two design choices follow from this:

1. **NLP only selects days.** I do *not* build a continuous war-risk index. A GDELT daily volume series (six war phrases) flags **H** (high-news) vs **L** (low-news) trading days; direction is used only for the three-regime extension, not for identification.
2. **Global variables.** The 2003 paper used nine U.S. variables; I use a global set (U.S. Treasuries + inflation breakeven, U.S. and European equities, Brent oil, gold, the dollar, and U.S. credit spreads), normalizing on the **2-year Treasury yield** exactly as the paper does.

2. Data

Variable	Source	Id	Units of daily change
2-Year Treasury Yield (anchor)	FRED	DGS2	percentage points
10-Year Treasury Yield	FRED	DGS10	percentage points
10Y Breakeven Inflation	FRED	T10YIE	percentage points
S&P 500	Yahoo	^GSPC	percent (log)
Euro Stoxx 50	Yahoo	^STOXX50E	percent (log)
Brent crude (front-month)	Yahoo	BZ=F	USD
Gold	Yahoo	GC=F	USD
U.S. Dollar Index	Yahoo	DX-Y.NYB	percent (log)
US High-Yield OAS	FRED	BAMLH0A0HYM2	percentage points
US BBB (IG) OAS	FRED	BAMLC0A4CBBB	percentage points

War news comes from the **GDELT DOC 2.0 API** (free, no key): daily article counts for `iran war`, `iran israel`, `iran strike`, `iran missile`, `iran nuclear`, `strait of hormuz`. High-news days are flagged as the top decile of the combined intensity (tunable with `--quantile` in step 02).

3. Method

3.1 NLP flagging

For each trading day I compute a war-news intensity (the max article count across the six war phrases, so a day is "high" if *any* war theme spikes). Days in the top decile of intensity are **H** (flag = 1); all other trading days are **L** (flag = 0). This is the professor's 1/0 flag. For the three-regime extension (§5), headlines on H days are classified with a war lexicon plus **FinBERT** (ProsusAI/finbert) sentiment into *bad* (escalation) / *good* (de-escalation).

3.2 Heteroskedasticity-based identification

For each pair (anchor $\Delta x_1 = 2Y$ yield, variable $\Delta x_2 = j$), the reduced form is $\Delta x = D \cdot z + \mu$ with the war-risk factor z_1 normalized to a unit impact on Δx_1 . The covariance matrix is $\Omega = D \Sigma z D' + \Sigma \eta$. Under the identifying assumptions (war risk orthogonal to other factors; only its variance shifts across H/L),

$$\Delta \Omega = \Omega_H - \Omega_L = \Delta \sigma^2(z_1) \cdot \begin{bmatrix} 1 & d \\ d & d^2 \end{bmatrix}$$

so the loading d of variable j on war risk is recovered two ways:

```
d_w1 = ΔΩ21 / ΔΩ11      (instrument built from the anchor)
d_w2 = ΔΩ22 / ΔΩ21      (instrument built from variable j)
d_w3 = inverse-variance-weighted combination
```

Variances are mean squares (mean of Δx^2 , mean of $\Delta x_1 \cdot \Delta x_2$), matching the paper's footnote 12. Standard errors are bootstrapped (resampling H and L separately, 3 000 draws), so the t-statistics are robust to the small H sample and to unequal H/L sizes. Reported coefficients are multiplied by **-0.25**, i.e. the effect of a war-risk *increase* large enough to move the 2Y yield **-25 bp**.

Table 3 reports the variance decomposition: $\text{predicted change} = d^2 \cdot \Delta \text{Var}(2Y)$, % on H days = $\text{pred} / \text{Var}_H(x_j)$, and % on all days = $n_H \cdot \text{pred} / (n_H \cdot \text{Var}_H + n_L \cdot \text{Var}_L)$ (a lower bound, assuming daily independence).

4. Results

All three tables are generated by `scripts/04_estimate.py` into `outputs/`. Two normalizations are reported, because the choice of anchor turns out to be the central empirical issue (§4.1).

4.1 The 2Y anchor fails for 2026 — a finding, not a bug

The paper normalizes the war-risk factor on the **2-year Treasury yield**, i.e. it reads the factor's variance shift off the 2Y yield's variance shift. That requires the 2Y yield's variance to **rise** on high-news days. In 2026 it does not:

Anchor	Var on H days	Var on L days	$\Delta \Omega_{11}$	identification
DGS2 (paper)	0.002393	0.002647	-0.000254	X fails
BRENT	63.13	12.77	+50.36	✓ holds

The war-news days (early March) were a *flight-to-quality* period in which Treasury yields were unusually **calm**; the yield volatility arrived later (April–September: inflation, debt, the Warsh Fed), and those days are the L sample. The 2003 "war risk $\Rightarrow$ Treasury volatility" channel is therefore absent in 2026 — exactly the professor's prior. Every *other* variable does have higher variance on H days (oil, equities, gold, FX, credit spreads).

I report the paper-faithful **2Y-anchored** version as a diagnostic (`outputs/table2_sensitivity.csv`) and the **Brent-anchored** version as the substantive estimate (`..._BRENT.csv`); Brent is also the more natural anchor for an Iran (energy-chokepoint) shock.

4.2 Table 1 — high war-news days (2026-02-28 → 2026-09-18)

14 trading days in the top decile of GDELT war-news volume, 25 English headlines each (outputs/table1_war_news_days.csv), all in early March plus 1 Apr and 8 Apr. Direction (war lexicon + FinBERT) classifies **13 as bad (escalation)** and **1 as good** (8 Apr, lex_net = -12: "US and Iran agree to a two-week ceasefire").

date	n	lex_net	regime	representative headline
2026-03-03	25	+3	bad	Iran and the US have been at war for decades. Here when it began
2026-03-04	25	+7	bad	Trump Declares War On Iran, But Reasons For War Keep Changing
2026-03-09	25	+1	bad	Mojtaba Khamenei is Iran's new supreme leader; oil prices breach \$100
2026-03-10	25	+13	bad	Iran IRGC Responds To Trump, Says It Will Determine The End Of War
2026-03-13	25	+8	bad	Iran claims US aircraft carrier Lincoln struck, Washington denies damage
2026-03-16	25	+8	bad	Dubai Airport suspends ops after drone strikes; Japan says no to Hormuz plan
2026-03-17	25	+1	bad	Top US counterterrorism official resigns over Iran war
2026-04-01	25	+1	bad	Why the US is at war with Iran
2026-04-08	25	-12	good	US and Iran agree to a two-week ceasefire

4.3 Table 2 — sensitivity to war risk (Brent anchor)

Normalized to a war-risk increase large enough to raise Brent by **\$10** (outputs/table2_sensitivity_BRENT.csv). d_w1 / d_w2 are the two just-identified estimators; d_w3 is the combined one.

Variable	d_w1	d_w2	d_w3	t (w3)
2-Year Treasury Yield (pp)	0.008	-0.064	0.008	0.10
10-Year Treasury Yield (pp)	-0.005	0.203	-0.005	0.11
10Y Breakeven Inflation (pp)	0.008	-0.005	0.008	0.37
S&P 500 (%)	-0.88	-1.11	-0.89	1.04
Euro Stoxx 50 (%)	-2.36	-2.97	-2.39	2.77
Gold (\$)	-94.9	-82.1	-86.8	0.84
U.S. Dollar Index (%)	0.39	0.59	0.55	1.47
US High-Yield OAS (pp)	0.115	0.118	0.117	3.11
US BBB OAS (pp)	0.023	0.033	0.026	1.31

Two effects are significant: a war-risk shock **lowers European equities** (-2.4%, t = 2.9) and **widens high-yield credit spreads** (+11.7 bp, t = 3.1); the dollar strengthens weakly (t = 1.5) while Treasury yields do not respond.

4.4 Table 3 — variance decomposition (Brent anchor)

outputs/table3_variance_BRENT.csv. pred = d²·ΔVar(anchor) is the variance attributable to the war-risk factor's extra volatility on H days.

Variable	Var on L	Var on H	pred change	% on H	% all
High-Yield OAS	0.0019	0.0087	0.0069	79.1	27.1
Dollar index	0.098	0.214	0.154	72.1	14.2
BBB OAS	0.0001	0.0005	0.0003	71.1	25.8
Euro Stoxx 50	0.845	4.373	2.880	65.9	24.7
Gold	5372	9295	3796	40.8	6.7

Variable	Var on L	Var on H	pred change	% on H	% all
S&P 500	0.697	1.191	0.398	33.4	5.4
10Y breakeven	0.00049	0.00047	0.00003	6.2	0.6
10Y Treasury	0.00206	0.00154	0.00001	0.8	0.1
2Y Treasury	0.00265	0.00239	0.00003	1.3	0.1
Brent (anchor)	12.77	63.13	—	—	—

War news accounts for roughly two-thirds to four-fifths of the H-day variance of credit spreads, the dollar and European equities, and ~14–27% of their variance over the whole window. For yields it is ~1%.

4.5 Verification (`scripts/05_verify.py`)

The whole chain was re-derived bottom-up from the raw files rather than trusted:

1. **Data integrity** — every change column is reproduced exactly by differencing its level column on the *anchor's* trading calendar. (This check originally failed for the HY OAS because FRED's series carries a few weekend/holiday rows — e.g. Sunday 31 May — which were being differenced against the wrong neighbour. Fixed by restricting each series to the anchor's calendar *before* differencing; the results then barely moved.)
2. **Rank-1 signature** — the single-factor model predicts $\Delta\Omega_{21}^2 = \Delta\Omega_{11} \cdot \Delta\Omega_{22}$, i.e. the ratio r should be ≈ 1 . Measured: $r = \mathbf{0.98}$ (HY OAS), **0.80** (Euro Stoxx), **0.79** (S&P 500), 1.16 (gold), 0.67 (dollar), 0.68 (BBB). The structure holds. Yields give $r = \text{nan}$ only because their $\Delta\Omega_{22} < 0$ (\$4.1).
3. **Recomputation** — `d_w1 / d_w2` recomputed by hand reproduce the saved table for all nine variables.
4. **Placebo** — the real H split's anchor-variance shift is **+50.36**; 5 000 random 14-day splits give mean +0.03, sd 10.4, max +56.8 (**p = 0.0002**). The NLP flag therefore selects genuinely high-variance days, not arbitrary ones.
5. **Threshold sensitivity** — from `--quantile 0.80` to 0.95 ($n_H = 27 \rightarrow 7$) the coefficients barely move: Euro Stoxx -2.4 to -3.0, HY OAS +0.106 to +0.134, dollar +0.41 to +0.59. The result is not an artefact of the cut.
6. **Variance-decomposition algebra** — `pred = d2·ΔVar(anchor)` and % on H reproduce Table 3 exactly.

Run it with `uv run python scripts/05_verify.py`.

How to read Table 2

- A **negative** equity coefficient means a war-risk shock *lowers* equities; a **positive** spread coefficient means spreads *widen*; a positive USD coefficient means the dollar *appreciates*.
- Closeness of `d_w1` and `d_w2` is the paper's informal check that the linear/homoskedastic structure is not badly violated (they agree well here).

5. Benchmark vs. the 2003 result (and the 2026 prior)

The professor's prior, checked against the Brent-anchored Table 2:

Variable	2003 (Iraq)	Expected 2026	Found (2026, Brent anchor)
Treasury yields	fell ~26 bp	rise	no response — 2Y $\Delta\Omega_{11} < 0$; $d \approx 0$, $t \approx 0.1$
Breakeven inflation	fell	ambiguous	+0.8 bp, $t = 0.4$ (nil)
Oil	rose	rise	anchor (by construction the shock raises Brent)
Credit spreads	widened	widen	widen — HY +11.7 bp (t = 3.1), BBB +2.6 bp ($t = 1.3$)
Equities	fell	fall	fall — Euro Stoxx 50 -2.4% (t = 2.9), S&P 500 -0.9% ($t = 1.0$)

Variable	2003 (Iraq)	Expected 2026	Found (2026, Brent anchor)
Gold	barely moved	muted	−\$87, t = 0.8 (nil)
Dollar	fell	—	+0.55%, t = 1.5 (appreciates weakly — opposite to 2003)

The 2026 pattern is therefore **oil** → **European equities and credit spreads, with no Treasury-yield channel** — consistent with the professor's prior that the 2003 flight-to-quality yield response would not repeat.

Three-regime extension — data-limited

Within H days, split by NLP direction (bad/good war news). The classification in `data/day_regimes.csv` returns **13 bad days and only 1 good day** (8 Apr ceasefire), because every other high-news day falls inside the March escalation. The "good war news" regime therefore has $n = 1$ and **cannot be estimated separately** — effectively only "bad war news vs no war news" is identified. This is a property of the episode (a hot-war window), not of the method, and is reported as a limitation rather than a result.

6. Is heteroskedasticity-based identification the best approach?

Asked and answered honestly (the professor's second deliverable).

Why it is appropriate here.

1. **The factor is unobservable.** War risk cannot be quantified ("today's war risk = 7.3"), so any approach that needs a measured level is out. Het-ID needs only *which days* had elevated war-news variance — exactly what NLP provides.
2. **It handles simultaneity without an instrument.** Asset prices and the war narrative are jointly determined; het-ID uses the *second-moment* shift as a probabilistic instrument, so no exclusion restriction is required.
3. **It is robust to regime misspecification.** Rigobon (2003, Prop. 3–4) shows the estimator stays consistent if the H/L split is imperfect (as long as the covariance matrices still differ in rank) — important when the day-flag comes from a noisy NLP classifier.
4. **No proxy = no attenuation.** A text-based war-risk index would be measured with error, biasing OLS loadings toward zero.

Why it is *not* obviously "best" for the 2026 exercise.

1. **Long-window confounding.** The paper's window was 10 weeks of *pre-war buildup*. Mine is ~6.5 months spanning war onset, an active phase, and a simmer-down — over which *other* factors' variances (Fed policy under Warsh, inflation/debt shocks, oil-supply shifts) also changed. If H = "hot-war March" and L = "calm September", the assumption that *only* war-risk variance shifts is strained, and d can pick up the March-vs-September macro shift. This is the central threat to validity in this exercise, and its empirical signature is visible in §4.1: the 2Y yield's variance *falls* on H days, so the paper's own anchor no longer identifies the factor.
2. **One factor, many things.** War news is multidimensional (likelihood, duration, energy disruption); collapsing it to one factor discards the direction and composition — the paper itself concedes this.
3. **Pairwise estimation.** I (like the paper) estimate two variables at a time for small-sample reasons, sacrificing the overidentifying restrictions a multivariate het-ID would give.
4. **Direction discarded.** The core method ignores whether news is good or bad, so it averages over both — the three-regime split is only a partial fix.

Alternatives, evaluated.

Alternative	Idea	Pros	Cons
Direct text index + OLS/IV	Regress returns on a daily war-risk surprise (e.g. Caldara–Iacoviello GPR, or a custom FinBERT war score)	Uses level <i>and</i> sign; simple; dynamic	Measurement error → attenuation; needs an "expected news" benchmark to get the <i>surprise</i>
Event study	Abnormal return around discrete war events	Standard, transparent, clean causality	Needs clean event dating; overlaps; no "per-unit-of-risk" loading
Structural VAR / sign restrictions	Identify a war shock via signs (oil↑, equities↓...)	Multivariate, dynamic IRFs	Only set-identified; signs must be assumed; not tied to actual news
High-frequency / intraday ID	Use minutes around a headline timestamp	Gold-standard causality (news = exogenous surprise)	Needs intraday data + precise timestamps; labor-intensive; not a daily factor loading
Multivariate het-ID (GMM)	N variables at once → overidentifying restrictions + parameter-stability test	Natural upgrade; testable	Needs more H days than we have

Verdict. For *replicating the paper* (the assignment's explicit ask) with NLP day-flagging, het-ID is the right and only directly-applicable method — and it is the only one that needs no quantification of war risk. But I would **not** call it unambiguously the best estimator of the *causal* 2026 sensitivities: the long-window confound means I would report it alongside (a) an event study on the discrete escalation dates and (b) a FinBERT/GPR *surprise* index regressed via IV, and treat intraday identification as the gold standard if the data were available. The het-ID numbers are best read as "the shift in second moments on war-news days", which is precisely what the professor asked for, and no more than that.

7. Limitations

- **Regime confound** (§6): war onset vs. post-war macro drift is not separable in a single H/L contrast over 6.5 months.
- **Flag threshold is arbitrary** (top decile by default); the H set and thus d move with `--quantile`. A robustness table across quantiles belongs in the final notebook.
- **NLP noise — and a multilingual surprise.** GDELT volume counts *mentions*, not uniquely war-relevant stories, and FinBERT is finance-tuned rather than geopolitics-tuned. More concretely: the first headline pull carried **no language filter** and came back **97% non-English** (210 headlines from 28 countries; only 7 from U.S./UK sources), which the English lexicon and FinBERT could not score — 3% lexicon hits, 10/14 days "unclear", FinBERT `fb_neg` ≈ 0.03 (everything read as neutral). Adding `source_lang:english` plus a client-side language guard produced **350 English headlines, 21% lexicon hits, and FinBERT `fb_neg` of 0.10–0.68**; the direction classification only became usable after that fix. This is the professor's "novel phrasing" warning in a broader form — a novel *vocabulary* and a *multilingual* corpus both degrade an English-centric pipeline. It also reinforces why het-ID suits this task: the core result needs only *which* days were news-heavy, not what the stories said.
- **Three-regime split is data-limited.** 13 of the 14 H days are escalation news, so the "good war news" regime has $n = 1$ and cannot be estimated (§5).
- **Sample size.** ~14–30 H days yields wide bootstrap standard errors; point estimates are more reliable than t-stats here.
- **Unequal H/L sizes.** The moment-based estimators handle this correctly, but the bootstrapped standard errors assume the H and L samples are each representative of their regime.